Felix Jimenez

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Summary

Applied Scientist and PhD Statistician specializing in foundation models, representation learning, and probabilistic inference. Experience spanning LLM post-training, vision foundation model pretraining, self-supervised learning, and scalable optimization for modern multimodal systems.

Experience

Applied Scientist — May 2025 – Present
Amazon — *Seattle, WA*

- Foundation model post-training—developed post-training pipelines for representation- and preference-aligned adaptation, including multimodal settings that treated embeddings as an input modality
- Vision foundation model pretraining—built distributed self-supervised pretraining and data-curation pipelines for representation learning and downstream model selection
- 3D generative world modeling—built a real-to-sim pipeline that generated grounded, simulation-ready 3D meshes from sparse real-world images using reconstruction, novel-view synthesis, Gaussian splatting, and meshing

Research Assistant — May 2020 – May 2025
University of Wisconsin—Madison — *Madison, WI*

- Scalable Bayesian optimization and uncertainty quantification—developed methods for optimization and uncertainty estimation in high-dimensional settings
- NASA OCO-2 atmospheric modeling—designed a neural system for real-time XCO_2 estimation from satellite spectra, with uncertainty quantification robust to forward-model error

Research Intern — May 2023 – Aug 2023
Toyota Research Institute — *Los Altos, CA*

- Self-supervised learning for battery diagnostics—developed self-supervised learning methods to train neural networks on experimental battery time-series data and learn reusable representations for downstream characterization and evaluation

Research Intern — May 2022 – Aug 2022
Microsoft Research — *Boston, MA*

- Bayesian optimization for model training—used gradient statistics to predict learning spikes and final training performance, enabling early experiment termination and faster hyperparameter search

Mathematical Statistician — May 2018 – Aug 2020
NIST — *Boulder, CO*

- Scientific machine learning for measurement systems—collaborated with domain scientists and engineers on physical measurement and inference problems, developing machine learning and statistical methods for representation learning, calibration, and estimation

Education

University of Wisconsin—Madison — Madison, WI
Ph.D. in Statistics — *2025*

University of Colorado Boulder — Boulder, CO
B.S./M.S. in Applied Mathematics — *2018*

Technical Skills

Programming: Python, PyTorch, JAX, CUDA, Distributed Training, Git, Bash

ML/AI: Foundation Models, Post-Training, Self-Supervised Learning, Bayesian Optimization, Uncertainty Quantification

Multimodal / 3D: Representation Learning, 3D Reconstruction, Novel-View Synthesis, Gaussian Splatting, Mesh Extraction

Systems: AWS, Docker, MLflow, vLLM, SGLang